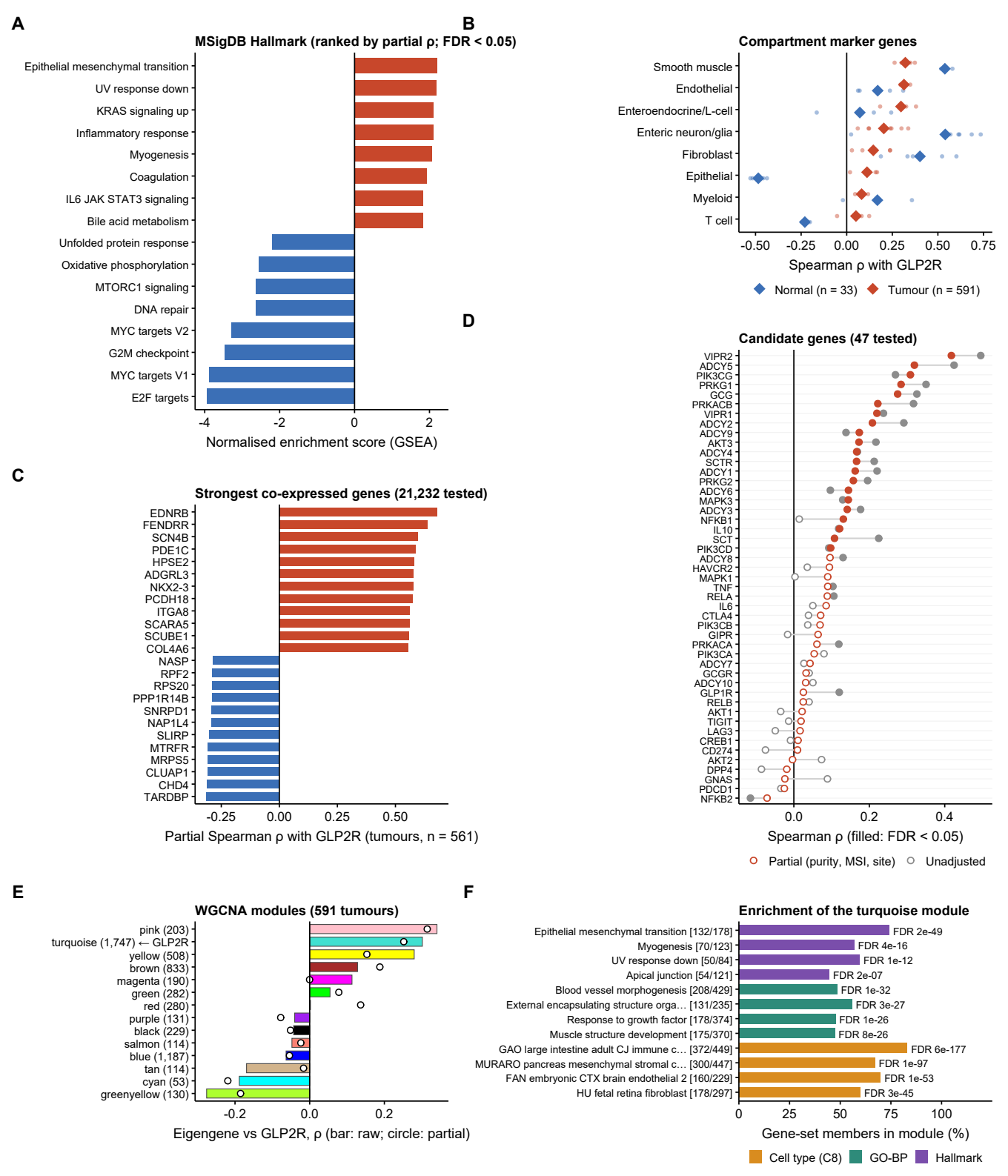

A
MSigDB Hallmark (ranked by partial ρ; FDR < 0.05)
Epithelial mesenchymal transition
UV response down
KRAS signaling up
Inflammatory response
Myogenesis
Coagulation
IL6 JAK STAT3 signaling
Bile acid metabolism
Unfolded protein response
Oxidative phosphorylation
MTORC1 signaling
DNA repair
MYC targets V2
G2M checkpoint
MYC targets V1
E2F targets
-4
-2
0
2
Normalised enrichment score (GSEA)
B
Compartment marker genes
Smooth muscle
Endothelial
Enteroendocrine/L-cell
Enteric neuron/glia
Fibroblast
Epithelial
Myeloid
T cell
-0.50
-0.25
0.00
0.25
0.50
0.75
Spearman ρ with GLP2R
Normal (n = 33)
Tumour (n = 591)
C
Strongest co-expressed genes (21,232 tested)
EDNRB
FENDRR
SCN4B
PDE1C
HPSE2
ADGRL3
NKX2-3
PCDH18
ITGA8
SCARA5
SCUBE1
COL4A6
NASP
RPF2
RPS20
PPP1R14B
SNRPD1
NAP1L4
SLIRP
MTRFR
MRPS5
CLUAP1
CHD4
TARDBP
-0.25
0.00
0.25
0.50
Partial Spearman ρ with GLP2R (tumours, n = 561)
D
Candidate genes (47 tested)
VIPR2
ADCY5
PIK3CG
PRKG1
GCG
PRKACB
VIPR1
ADCY2
ADCY9
AKT3
ADCY4
SCTR
ADCY1
PRKG2
ADCY6
MAPK3
ADCY3
NFKB1
IL10
SCT
PIK3CD
ADCY8
HAVCR2
MAPK1
TNF
RELA
IL6
CTLA4
PIK3CB
GIPR
PRKACA
PIK3CA
ADCY7
GCGR
ADCY10
GLP1R
RELB
AKT1
TIGIT
LAG3
CREB1
CD274
AKT2
DPP4
GNAS
PDCD1
NFKB2
0.0
0.2
0.4
Spearman ρ (filled: FDR < 0.05)
Partial (purity, MSI, site)
Unadjusted
E
WGCNA modules (591 tumours)
pink (203)
turquoise (1,747) ← GLP2R
yellow (508)
brown (833)
magenta (190)
green (282)
red (280)
purple (131)
black (229)
salmon (114)
blue (1,187)
tan (114)
cyan (53)
greenyellow (130)
-0.2
0.0
0.2
Eigengene vs GLP2R, ρ (bar: raw; circle: partial)
F
Enrichment of the turquoise module
Epithelial mesenchymal transition [132/178]
Myogenesis [70/123]
UV response down [50/84]
Apical junction [54/121]
Blood vessel morphogenesis [208/429]
External encapsulating structure orga… [131/235]
Response to growth factor [178/374]
Muscle structure development [175/370]
GAO large intestine adult CJ immune c… [372/449]
MURARO pancreas mesenchymal stromal c… [300/447]
FAN embryonic CTX brain endothelial 2 [160/229]
HU fetal retina fibroblast [178/297]
FDR 2e-49
FDR 4e-16
FDR 1e-12
FDR 2e-07
FDR 1e-32
FDR 3e-27
FDR 1e-26
FDR 8e-26
FDR 6e-177
FDR 1e-97
FDR 1e-53
FDR 3e-45
0
25
50
75
100
Gene-set members in module (%)
Cell type (C8)
GO-BP
Hallmark